

- 14 C** coin loses contact when piston retracts downwards at an acceleration larger than magnitude of free fall acceleration i.e.

$$|a_{\text{SHM}}| \geq |g|$$

$$(\omega^2)x_0 \geq g$$

$$= (2\pi f)^2 x_0$$

$$f = \sqrt{\frac{9.81}{0.07} \frac{1}{4\pi^2}}$$

$$= 1.9 \text{ Hz}$$

15 D $\lambda = 2(0.90) = 1.80 \text{ m}$